

AI-BFSI Internship Project Documentation

AI Prompt Logs – Subscription Waste Detector

Executive Summary

Generative AI was used as a development assistant throughout the implementation of the Subscription Waste Detector. AI supported activities such as code generation, debugging, documentation, interface improvements, requirement analysis, and report preparation.

The AI system was used as a **productivity tool** rather than an autonomous developer. All generated responses were reviewed, modified, tested, and validated before being incorporated into the project.

Purpose of AI Usage

AI was used to:

- Accelerate software development
- Improve code quality
- Assist with debugging
- Generate technical documentation
- Refine user interface design
- Improve report writing
- Explain programming concepts
- Assist with testing and validation

Final implementation decisions remained under developer control.

AI Tools Used

AI Tool	Purpose
ChatGPT	Development assistance, debugging, documentation, report generation, code explanation
Groq API	Generate AI-powered subscription cost-saving recommendations within the application

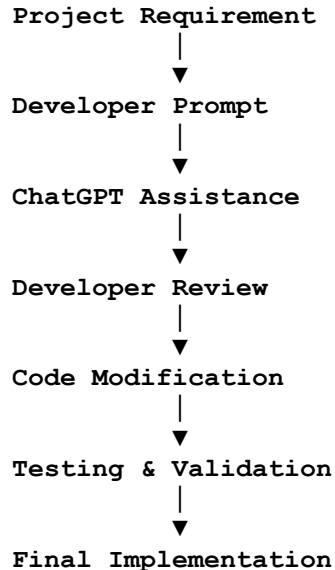
Note: ChatGPT was used during development, while the Groq API is integrated into the final application to provide AI recommendations to end users.

Representative Prompt Log

Stage	Example Prompt	Purpose	Outcome
Project Planning	"Suggest suitable AI-based BFSI project ideas."	Idea generation	Selected Subscription Waste Detector

Requirement Analysis	"Help define functional requirements for a subscription analysis application."	Requirement definition	Requirements documented
Architecture Design	"Design a modular architecture for a Streamlit application."	Architecture planning	Modular solution architecture created
Development	"Implement CSV and Excel upload functionality using Streamlit and Pandas."	Feature implementation	File upload module completed
Development	"Generate recurring subscription detection logic."	Core functionality	Subscription detection implemented
Dashboard	"Create Plotly visualizations for subscription spending."	Dashboard development	Interactive charts created
AI Integration	"Integrate Groq API to generate subscription optimization recommendations."	AI integration	Recommendation module implemented
Debugging	"Identify and resolve errors in Streamlit application."	Troubleshooting	Runtime issues resolved
Documentation	"Generate software engineering documentation for the implemented project."	Documentation	Technical reports prepared
UI Improvement	"Improve report layout and professional presentation."	Documentation refinement	Final report formatting completed

AI-Assisted Development Workflow



This workflow ensured that AI-assisted suggestions were always reviewed and validated before being incorporated into the project.

Development Activities Assisted by AI

Planning

- Brainstorming project ideas

- Refining project objectives
- Identifying project scope

Programming

- Python code assistance
- Streamlit interface development
- Data processing logic
- Subscription detection implementation

Debugging

- Runtime error analysis
- Logic corrections
- Code optimization
- Import and dependency troubleshooting

Documentation

- Software engineering reports
- Project proposal
- Architecture documentation
- Requirements documentation
- Testing documentation

Report Preparation

- Professional formatting
- Technical writing refinement
- Academic documentation
- Software lifecycle documentation

AI Usage Principles

- AI-generated responses were reviewed before implementation
- Generated code was tested within the application
- Incorrect or unsuitable suggestions were modified or discarded
- Documentation was aligned with actual implemented functionality
- AI recommendations were treated as assistance, not final implementation

Benefits of AI Assistance

- Faster problem solving
- Reduced development time
- Improved documentation quality
- Better understanding of technical concepts
- Increased productivity
- Faster debugging
- More structured project documentation

Limitations of AI

- Generated code occasionally required modification
- Suggested implementations required testing before adoption
- Documentation required verification to ensure alignment
- Final technical decisions remained the responsibility of the developer

Compliance Statement

The Subscription Waste Detector was designed, implemented, tested, and documented under developer supervision. AI was used as an assistive tool to improve productivity and documentation quality.

All implementation decisions, software validation, and final project outcomes were verified before inclusion in the final application and documentation.

Conclusion

Artificial Intelligence played a valuable supporting role throughout the development of the Subscription Waste Detector by assisting with planning, implementation, debugging, documentation, and report preparation.

Rather than replacing the software development process, AI enhanced productivity while preserving human oversight and technical validation.